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United States Patent Application Publication

20250273546

Kind Code

A1

Publication Date

August 28, 2025

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INTEGRATED CIRCUIT PACKAGING WITH INSULATION STRUCTURE TO CONTROL GAP HEIGHT AND DIE TILT

Abstract

A current sensor integrated circuit (IC) package includes an insulation structure disposed between a semiconductor die and a lead frame to control gap height and prevent the die from tilting and dropping the magnetic field coupling between the die the primary conductor. An insulation structure is disposed between the die and the lead frame such that the die remains level and magnetic coupling remains intact. An insulation structure may control the gap height between the lead frame and the die evenly during transfer molding, by supporting the die across its length and/or width. Epoxy dots are also or instead used to control the gap height and eliminate die tilt.

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Appl. No.: 18/583987

Filed: February 22, 2024

Publication Classification

Int. Cl.: H01L23/495 (20060101); H01L23/00 (20060101); H01L25/065 (20230101)

U.S. Cl.:

CPC H01L23/49575 (20130101); H01L23/4951 (20130101); H01L23/49558 (20130101);
H01L24/05 (20130101); H01L24/13 (20130101); H01L25/0657 (20130101);
H01L2224/05005 (20130101); H01L2224/05022 (20130101); H01L2224/13005

Background/Summary

BACKGROUND

[0001] During the manufacture of some current sensor integrated circuit (IC) packages, a semiconductor die carrying the sensing elements is disposed over a lead frame that includes the primary conductor and a number of signal leads. During assembly, placement of the die over an insulation material creates a gap between the die, the primary conductor and the signal leads. A gap height differential may occur when, for example, the gap between the semiconductor die and the primary conductor differs from the gap between the semiconductor die and the signal leads. The differing gap heights may be a result of inconsistent or collapsed solder bumps, isolation tape thickness, and/or a combination of both.

[0002] For example, changing the thickness of an isolation tape on the primary conductor may require increasing the size of solder bumps on the signal leads. Doing so, however, may cause the solder bumps to collapse and the die to tilt towards the signal leads. The tilt can compromise the magnetic field coupling to the lead frame. In some cases, the die may undesirably touch the lead frame due to a substantial tilt.

[0003] Die tilt may cause partial discharge failures on voltage isolation testing as well as sensitivity drift. FIG. 1 depicts a partial view of a prior art IC package **100** with an undesirable die tilt. The IC package **100** includes a die **102** partially disposed over an insulation material **108**, a primary conductor **104** and signal leads **106**. As shown, the placement of the die **102** over a gap **110** between the primary conductor **104** and the signal leads **106** causes the die **102** to be disposed at an angle with respect to the primary conductor **104** and the signal leads **106**. This tilt compromises the coupling of the die **102** with the primary conductor **104** and the signal leads **106** and may cause the IC package to fail.

SUMMARY

[0004] Aspects of the present disclosure relate to current sensor integrated circuits (IC) including an insulation structure disposed between a semiconductor die and a lead frame to control gap height and/or prevent the die from tilting and degrading the magnetic field coupling between the die the primary conductor. An insulation structure may be disposed between the die and the lead frame such that the die remains level and magnetic coupling remains intact. According to one aspect, an insulation structure may control the gap height between the lead frame and the die evenly during transfer molding by supporting the die across its length and/or width. According to other aspects, epoxy dots may also or instead be used to control the gap height and reduce die tilt.

[0005] According to one aspect, a current sensor IC package may include a lead frame having a primary conductor and signal leads. The lead frame may have a lead frame surface. A semiconductor die may have a die surface adjacent to the lead frame surface and may include a magnetic field sensing element supported by the semiconductor die. The magnetic field sensing element may be configured to sense a magnetic field associated with a current through the primary conductor and generate an output signal indicative of the current. An insulation structure may have first and second portions. The first portion may be located between a first location on the die surface and the primary conductor, and the second portion may be located between a second location on the die surface and a first one of the signal leads. The insulation structure may be configured to prevent a tilt of the semiconductor die.

[0006] The current sensor IC package can include one or more of the following features alone or in combination. The first and second portions of the insulation structure may comprise a single body

of material. The single body may comprise a T-shape. The insulation structure may comprise a polymer film. The insulation structure may further comprise an adhesive layer. The polymer film and the adhesive layer may be provided as a tape. The tape may extend beyond a periphery of the semiconductor die. The tape may extend beyond the primary conductor adjacent to the signal leads by at least 0.4 mm. The insulation structure may comprise one or more epoxy dots. The first portion may comprise a polymer film and the second portion may comprise an epoxy dot. The epoxy dot may be substantially rectangular, substantially oval, or polygonal. The epoxy dot may be screen printed onto the first one of the signal leads. The epoxy dot may be dimensioned to be within dimensions of the first one of the signal leads. The insulation structure may be adapted to support the semiconductor die across one or both of a width or a length of the semiconductor die.

[0007] According to another aspect, a method of manufacturing a current sensor IC package may include providing a lead frame comprising a primary conductor and signal leads. The lead frame may have a lead frame surface. A semiconductor die having a die surface adjacent to the lead frame surface may be positioned such that the semiconductor die supports a magnetic field sensing element. The magnetic field sensing element may be configured to sense a magnetic field associated with a current through the primary conductor and to generate an output signal indicative of the current. A first portion of an insulation structure may be positioned between a first location on the die surface and the primary conductor. A second portion of the insulation structure may be positioned between a second location on the die surface and a first one of the signal leads. The insulation structure may be configured to prevent a tilt of the semiconductor die.

[0008] The method can include one or more of the following features alone or in combination. The first and second portions of the insulation structure may comprise a single body of material.

[0009] The single body may comprise a T-shape. The insulation structure may comprise a polymer film. The insulation structure may further comprise an adhesive layer. The polymer film and the adhesive layer may be provided as a tape. The tape may extend beyond a periphery of the semiconductor die. The tape may extend beyond the primary conductor adjacent to the signal leads by at least 0.4 mm. The insulation structure may comprise one or more epoxy dots. The first portion may comprise a polymer film and the second portion may comprise an epoxy dot. The epoxy dot may be substantially rectangular, substantially oval, or polygonal. Positioning the epoxy dot may comprise screen printing the epoxy dot onto the first one of the signal leads. The epoxy dot may be dimensioned to be within dimensions of the first one of the signal leads. The insulation structure may be adapted to support the semiconductor die across one or both of a width or a length of the semiconductor die.

[0010] According to another aspect, a current sensor IC package may include a lead frame having a primary conductor and signal leads. The lead frame may have a lead frame surface. A semiconductor die may have a die surface adjacent to the lead frame surface and may include a magnetic field sensing element supported by the semiconductor die. The magnetic field sensing element may be configured to sense a magnetic field associated with a current through the primary conductor and generate an output signal indicative of the current. An insulation structure may have first and second portions. The first portion may be located between a first location on the die surface and the primary conductor, and the second portion may be located between a second location on the die surface and a first one of the signal leads. The insulation structure may be configured to form a first gap height between the first location on the die surface and the primary conductor and a second gap height between the second location on the die surface and the first one of the signal leads.

[0011] The current sensor IC package can include the first gap height and the second gap height being substantially equal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The manner and process of making and using the disclosed embodiments may be appreciated by reference to the figures of the accompanying drawings. It should be appreciated that the components and structures illustrated in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principals of the concepts described herein. Like reference numerals designate corresponding parts throughout the different views. Furthermore, aspects of the present disclosure are illustrated by way of example and not limitation in the figures, in which:

[0013] FIG. 1 is a partial side-view of a prior art integrated circuit (IC) package including a semiconductor die disposed at a tilt.

[0014] FIG. 2A is a partially transparent top view of an IC package including an insulation structure according to aspects of the disclosure.

[0015] FIG. 2B is a partially transparent isometric view of the IC package in FIG. 2A according to aspects of the disclosure.

[0016] FIG. 2C is a partially transparent side view of the IC package in FIG. 2A according to aspects of the disclosure.

[0017] FIG. 3A is a top plan view of an IC package including a multi-portion insulation structure according to aspects of the disclosure.

[0018] FIG. 3B is a side view of the IC package of FIG. 3A according to aspects of the disclosure.

[0019] FIG. 3C is a top perspective view of the IC package of FIG. 3A according to aspects of the disclosure.

[0020] FIG. 4 is a partial top plan view of an IC package including an epoxy-based insulation structure according to aspects of the disclosure.

[0021] FIG. 5A is a top perspective view of an IC package including an epoxy-based insulation structure according to aspects of the disclosure.

[0022] FIG. 5B is a side view of the IC package of FIG. 5A according to aspects of the disclosure.

[0023] FIG. 6A is a partial top plan view of an IC package including shaped epoxy insulation structures according to aspects of the disclosure.

[0024] FIG. 6B is a partial top plan view of an IC package including alternative shaped epoxy insulation structures according to aspects of the disclosure.

[0025] FIG. 6C is a partial top plan view of an IC package including other alternative shaped epoxy insulation structures according to aspects of the disclosure.

[0026] FIG. 7A is a partial top plan view of an IC package including an epoxy insulation structure on a primary conductor according to aspects of the disclosure.

[0027] FIG. 7B is a top perspective view of the IC package of FIG. 7A according to aspects of the disclosure.

[0028] FIG. 7C is side view of the IC package of FIG. 7A according to aspects of the disclosure.

DETAILED DESCRIPTION

[0029] Referring to the various views of FIGS. 2A-2C, a current sensor integrated circuit (IC) package **200** includes a lead frame **205** with a primary conductor **204** and a plurality of signal leads, collectively labeled **212**. It will be appreciated by those of ordinary skill in the art that although the lead frame **205** is shown to have eight signal leads, other numbers, dimensions, spacing, and configurations of leads are possible.

[0030] A semiconductor die **206** may include a first magnetic field sensing element (not shown) for sensing a magnetic field associated with a current through the primary conductor **204** to generate a first output signal indicative of the current for coupling to a signal lead **212**. With this arrangement, the semiconductor die **206** is arranged in a so-called flip-chip configuration in which the sensing element is supported by a die surface proximal to the lead frame **205**.

[0031] According to one aspect, the magnetic field sensing element can be a single element or can

include more than one element, such as a dual Hall element or a quad Hall element or one or more magnetoresistance elements as are sometimes arranged in a bridge configuration and as may be used to implement differential magnetic field sensing.

[0032] As used herein, the term “magnetic field sensing element” is used to describe a variety of electronic elements that can sense a magnetic field. The magnetic field sensing element can be, but is not limited to, a Hall effect element, a magnetoresistance element, or a magnetotransistor. As is known, there are different types of Hall effect elements, for example, a planar Hall element, a vertical Hall element, and a Circular Vertical Hall (CVH) element. As is also known, there are different types of magnetoresistance elements, for example, a semiconductor magnetoresistance element such as Indium Antimonide (InSb), a giant magnetoresistance (GMR) element, for example, a spin valve, an anisotropic magnetoresistance element (AMR), a tunneling magnetoresistance (TMR) element, and a magnetic tunnel junction (MTJ). The magnetic field sensing element may be a single element or, alternatively, may include two or more magnetic field sensing elements arranged in various configurations, e.g., a half bridge or full (Wheatstone) bridge. Depending on the device type and other application requirements, the magnetic field sensing element may be a device made of a type IV semiconductor material such as Silicon (Si) or Germanium (Ge), or a type III-V semiconductor material like Gallium-Arsenide (GaAs) or an Indium compound, e.g., Indium-Antimonide (InSb).

[0033] The primary conductor **204** may have various shapes and dimensions to accommodate a range of current levels sought to be detected by the current sensor IC package **200** and the desired package footprint. In general, primary conductor **204** may include an input portion **214a**, an output portion **214b**, and a thinned current path region **214c** between the input and output portions. In use, a current flow may be established through the primary conductor **204** between the input and output portions **214a**, **214b**.

[0034] The input and output portions **214a**, **214b** of the primary conductor **204** may have respective terminal ends in the form of leads, collectively labeled **214**, configured for electrical connection to a printed circuit board (PCB) or other substrate on which the current sensor IC package **200** may be mounted. The leads **214** of the primary conductor **204** and signal leads **212** can take various forms, such as the illustrated leads that are bent to facilitate surface mount solder connection to a PCB or other substrate. The current sensor IC package **200** may be considered an SOIC (Small Outline Integrated Circuit) package. In other embodiments, package types include QFN (Quad-Flat No-Leads), DFN (Dual-Flat No-Leads), and the like.

[0035] The thinned current path region **214c** of the primary conductor **204** can have various shapes and other characteristics. Here, the thinned current path region **214c** is curved in a “horseshoe” shape to form a notch **216**. Placement of the magnetic field sensing element relative to the curved, thinned current path region **214c** and notch **216** may concentrate the magnetic field generated by current through the primary conductor **214** on the sensing elements. It will be appreciated that other shapes, dimensions, notches, and sensing element placement can be implemented to achieve concentration of the magnetic field.

[0036] According to one aspect, the magnetic field sensing element can include one or more [0037] elements that are substantially vertically aligned with a first side of the notch **216** and one or more elements that are substantially vertically aligned with a second side of the notch **216**. As current flows through the primary conductor **204**, having a magnetic field sensing element positioned to the sides of the conductor (rather than directly over or under the conductor) results in magnetic field with components perpendicular to the semiconductor die **206** such that the sensing elements may be planar Hall effect elements. In other aspects, the magnetic field sensing element or elements may be positioned directly over the primary conductor **204** to sense magnetic field components parallel to the surface of semiconductor die **206**, in which case sensing elements such as a vertical Hall element, a GMR, TMR, or AMR element may be used.

[0038] The lead frame **205** may comprise any suitable conductive material, such as copper or

copper alloy or Aluminum, and its features can be formed by various methods such as stamping or etching.

[0039] According to one aspect, the flip-chip configured semiconductor die **206** may be electrically and mechanically coupled to the lead frame **205** by various mechanisms, including bumps **210**, which may be or include solder bumps or copper pillars, as non-limiting examples, with which semiconductor die **206** is connected to leads **212**. The solder may be a Sn-Pb solder or a lead less solder, such as an Indium solder. One or more bumps **210** may be so-called “dummy” bumps, in that they provide only mechanical attachment and/or stability rather than both mechanical and electrical coupling. In the case of a dummy bump, the bond pad to which the dummy bump is connected is not electrically connected to the circuitry on the die **206**. Such electrical isolation may be achieved with materials including, but not limited to an oxide, a nitride, or a polymer isolation layer or combinations thereof.

[0040] In applications in which the primary conductor **204** can be at a relatively high voltage, safety specifications require that a certain electrical isolation be maintained between the primary conductor **204** and other parts of the circuitry (e.g., signal leads **212** coupled to an external system to which a sensor output signal is communicated). According to one aspect of the disclosure, electrical isolation between the primary conductor **204** and the flip-chip semiconductor die **206** may be achieved with various mechanisms, such as with an insulation structure **208** disposed, at least, between the primary conductor **204** and the proximal surface of the semiconductor die **206**.

[0041] According to one aspect, the insulation structure **208** may, in addition to providing electrical isolation between the primary conductor **204** and the semiconductor die **206**, may provide gap height control between the die **206** and the lead frame **205** and prevent die tilt during the manufacturing process. The insulation structure **208** may include or one or more portions adapted to support the semiconductor die **206** across its length and/or width such that the semiconductor die **206** is disposed flatly, or evenly, over the lead frame **205** reducing any die tilt. As used herein, zero die tilt refers to a gap height, for example between about 1.8 mm and 2.8 mm, between the lead frame **205** and the semiconductor die **206** that is substantially the same at the primary conductor **204** as it is at the signal leads **212**.

[0042] According to one aspect, a first portion **208a** of the isolation structure may be disposed between the semiconductor die **206** and the primary conductor and a second portion **208b** may be located between the semiconductor die **206** and at least one of the signal leads **212**. The first portion **208a** may form or create a first gap height h_1 between the semiconductor die **206** and the primary conductor **204**. The second portion **208b** may form or create a second gap height h_2 between the semiconductor die **206** and at least one of the signal leads **212**. According to one aspect, the insulation structure **208** may be substantially T-shaped, with a base portion, such as the first portion **208a**, disposed between the semiconductor die **206** and the current path region **214c** of the primary conductor **204** and a top portion, such as the second portion **208b**, disposed between the semiconductor die **206** and at least one of the signal leads **212**. Accordingly, the insulation structure **208** may support the semiconductor die **206** across the length and/or width of the semiconductor die **206**, such that the first gap height h_1 and the second gap height h_2 are substantially equal. In this way, the insulation structure **208** provides an even surface to which the semiconductor die **206** may be coupled or bonded with reduced or zero tilt.

[0043] According to one aspect, the base portion, such as first portion **208a**, of the insulation structure **208** may extend beyond the periphery of the semiconductor die **206** and/or the primary conductor **204** so as to achieve a required clearance specification. For example, in certain aspects, the insulation structure **208** may extend beyond the die **206** and/or primary conductor **204** by a distance of at least 0.4 mm. The overall size and shape of the insulation structure **208**, as further described herein, may vary with the size/shape of the die **206** and primary conductor **204**.

[0044] The insulation structure **208** may include an organic polymer such as polyimide, or an oxide insulating material like silicon dioxide in the form of a glass sheet, silicone or ceramic. For

example, the insulation structure **208** may be a polymer film, as may be provided in the form of a polyimide or Kapton® tape, as non-limiting examples. The insulation structure **208** may include an adhesive layer, in which case the polymer film and the adhesive layer may be provided as a tape with which the insulating layers are attached to the lead frame **205** and such an adhesive layer can itself provide insulation.

[0045] An attachment material can be provided to mechanically attach the semiconductor die **206** to the insulation structure **208**. According to some aspects, the semiconductor die **206** may be attached to the insulation structure **208** by an adhesive layer of the insulation structure **208** itself (i.e., an adhesive layer of an insulating tape). According to other aspects, semiconductor die **206** may be attached to the insulation structure **208** by a non-conductive adhesive, such as wafer backside coating or dispensed epoxy or a combination of both wafer backside coating and dispensed non-conductive epoxy, as non-limiting examples.

[0046] The current sensor IC package **200** may include insulating material in the form of a mold material **202** configured to encapsulate the semiconductor die **206** and portions of the lead frame **205**. The mold material **202** is shown to illustrate elements encapsulated within the IC package **200**. Various materials can be used to form the mold material **202**, including, but not limited to a plastic material.

[0047] Portions of the lead frame **205** may include features configured to enhance adhesion of the mold material **202** to the lead frame **205**, thereby serving as a “locking mechanism” to secure parts of the IC package **200** together. Here, holes **205a** through the primary conductor **204** and the signal leads **212** can provide such a locking mechanism.

[0048] According to one aspect of the disclosure, fabrication of the current sensor IC package **200** can be accomplished by first attaching the flip-chip semiconductor die **206** to the surface of the lead frame **205**. This step can include attaching the insulation structure **208** to the primary conductor **204** and at least one of signal leads **212**, then attaching the bumps **210**, such as soldering bumps or copper pillars, to plated areas of the surface of the lead frame **205** for electrical connection to the signal leads **212**.

[0049] The entire structure, excluding terminal ends of the signal leads **212** and primary leads **214**, can be overmolded with mold material **202** in a mold process step, following which terminal ends of the primary conductor leads **214** and the signal leads **212** can be bent, as shown. The mold material **202** may be formed by a transfer mold process, which may include one or more mold process steps.

[0050] Referring now to FIGS. 3A-3C, a current sensor IC package **300** is shown. The IC package **300** is similar to the current sensor IC package **200** of FIGS. 2A-2C, however with an alternative insulation structure **308**. Elements previously described with like reference numerals designate corresponding parts throughout the different views.

[0051] The current sensor IC package **300** may include an insulation structure **308** formed from a first portion **308a** and a second portion **308b** as separate bodies. According to one aspect, the insulation structure **308** may include a first portion **308a** disposed between the semiconductor die **206** and the primary conductor **204**. The insulation structure **308** may further include the second portion **308b** as a separate body from the first portion **308a** and disposed between the semiconductor die **206** and at least one of the signal leads **212**. The two-body insulation structure **308** may evenly support the semiconductor die **206** across its length and/or width to provide a consistent gap height and reduce any die tilt.

[0052] Referring now to FIG. 4, a current sensor IC package **400** may include an alternative insulation structure **408**. According to one aspect, the current sensor IC package **400** is shown as a DEN package, however other package types are possible, including QFN or SOIC. To this end, the terminal ends of primary conductor **404** may be conductive pads (not shown) exposed from side and bottom surfaces of the package **400** and signal leads **412** may likewise be or include conductive pads **413** exposed from side and bottom surfaces of the package **400**.

[0053] In other respects, the DFN package of FIG. 4 may be substantially similar as the SOIC package of FIGS. 3A-3C, including a molding material **402**, a primary conductor **404**, a semiconductor die **206** with one or more magnetic field sensing elements, bumps **410**, and magnetic field sensing elements **416**, the form and function of each being substantially similar to those previously described.

[0054] According to one aspect the insulation structure **408** may include or be formed by a first portion **408a** and one or more epoxy dots **408b** adapted to control gap height (i.e., the distance between the die **406** and the lead frame components) and reduce die tilt. According to one aspect, the insulation structure **408** may include a first portion **408a** similar to those previously described in connection with FIGS. 3A-3C, such as a polyimide tape, or the like. A second portion of the insulation structure **408** may include one or more epoxy dots **408b** disposed between one or more signal leads **412**. The epoxy dots **408b** may include non-conductive material such as 84-3 epoxy material made by Henkel Corporation, or the like. The epoxy dots **408b** may be disposed on the signal leads **412** using a screen-printing process. The epoxy dots **408b** may be sized and positioned to remain within the dimensions of the signal lead **412**. According to one aspect, the epoxy dots **408b** may be further dimensioned to a height or thickness substantially equal to the height or thickness of the first portion **408a** of the insulation structure **408**. The epoxy dots **408b**, according to one aspect, may be further dimensioned to be within the dimensions of the semiconductor die **206**. A mask may be used on the lead frame **405** exposing the target areas for the epoxy deposits to ensure the epoxy stays within the desired dimensions and geometry. Accordingly, the semiconductor die **206** may be disposed on the primary conductor **404** and the signal leads **412** with a controlled gap height and substantially no die tilt.

[0055] Referring now to the various views shown in FIGS. 5A-5B, a current sensor IC package **500** may be a similar SOIC package like that shown in FIGS. 2A-2C, and include a molding material **202**, a lead frame **205** with a primary conductor **204** and signal leads **212**, primary conductor terminal ends **214**, a semiconductor die **206**, and bumps **210** similar to those previously described herein. The insulation structure **508**, however, may be substantially similar to the insulation structure **408** of the IC package shown in FIG. 4. For example, the insulation structure **508** may include a first portion **508a** including or formed from a polyimide tape disposed between a current path region of the primary conductor **204** and the semiconductor die **206**.

[0056] The insulation structure **508** may further include a second portion including or formed from one or more non-conductive epoxy dots **508b** between at least one of the signal leads **212** and the semiconductor die **206**. The epoxy dots **508b** may include non-conductive material such as 84-3 epoxy material made by Henkel Corporation, or the like. The epoxy dots **508b** may be disposed on at least one of the signal leads **212** using a screen-printing process with a mask. The epoxy dots **508b** may be sized and positioned to remain within the dimensions of the signal lead **212**.

According to one aspect, the epoxy dots **508b** may be further dimensioned to a height or thickness substantially equal to the height or thickness of the first portion **508a** of the insulation structure **508**. Accordingly, the semiconductor die **206** may be disposed on the primary conductor **204** and the signal leads **212** with a controlled gap height and substantially no die tilt.

[0057] FIGS. 6A-6C show various configurations and shapes of an insulation structure **608** of a current sensor IC package **600** according to aspects of the disclosure. The current sensor IC package **600** may be a DFN package like that shown in FIG. 4, however other package types, including QFN or SOIC packages are possible. The current sensor IC package **600** may include, as previously described herein, a lead frame **405** with a primary conductor **404** and signal leads **412**. The signal leads **412** may include conductive pads **413**. The current sensor IC package **600** may also include a semiconductor die **406**, bumps **410**, magnetic field sensing elements **416**, and molding material **402**, as previously described.

[0058] According to one aspect, an insulation structure **608** may include a first portion **608a** disposed between the primary conductor **404** and the semiconductor die **406**. The first portion **608a**

may be substantially similar to the second portions **308a**, **408a**, **508a**, previously described. According to one aspect, a second portion may include one or more epoxy dots **609a** (FIG. **6A**), **609b** (FIG. **6B**), **609c** (FIG. **6C**) disposed between at least one of the signal leads **412** and the semiconductor die **406**.

[0059] FIG. **6A** depicts a second portion of the insulation structure **608** including rectangularly shaped epoxy dots **609a**. According to one aspect, the rectangular epoxy dots **609a** may be sized and positioned to extend beyond the dimensions of the semiconductor die **406** but still remain within the dimensions of the signal leads **412**. Such a configuration may provide added stability to the semiconductor die **406** while still providing substantial control over the gap height and die tilt. According to one aspect, the shapes of the epoxy dots **609a** may be varied within lead frame space, as described herein, to increase bonding without compromising additional free space or other connections.

[0060] FIG. **6B** depicts a second portion of the insulation structure **608** including oval shaped epoxy dots **609b**. According to one aspect, the oval epoxy dots **609a** may be sized and positioned to extend beyond the dimensions of the semiconductor die **406** but still remain within the dimensions of the signal leads **412**. Such a configuration may provide added stability to the semiconductor die **406** while still providing substantial control over the gap height and die tilt.

[0061] FIG. **6C** depicts a second portion of the insulation structure **608** including polygonal epoxy dots **609c**. According to one aspect, the polygonal epoxy dots **609c** may be sized and positioned to extend beyond the dimensions of the semiconductor die **406** and share a similar geometry profile as the signal lead **412** and still remain within the dimensions of the signal leads **412**. Such a configuration may provide added stability to the semiconductor die **406** while still providing substantial control over the gap height and die tilt.

[0062] According to another aspect of the disclosure, die tilt and gap height between a semiconductor die and a lead frame may be controlled using one or more epoxy dots on a primary conductor. Referring to the various views in FIGS. **7A-7C**, a current sensor IC package **700** with an alternative insulation structure **708** may be used to control gap height and reduce die tilt. The package **700** may be a SOIC, DEN, QFN, or other package type. The current sensor IC package **700** may include, as previously described herein, a primary conductor **704**, a semiconductor die **706** with at least one magnetic field sensing element **716**, signal leads **712**, bumps **710**, and molding material **702**. An insulation structure **708** may include one or more non-conductive epoxy dots **709** disposed on a current path region **704a** of the primary conductor **704**.

[0063] The epoxy dots **709** may be dimensioned to control the gap height between the primary conductor **704** and the semiconductor die **706**, thus substantially reducing any tilt of the die **706** that may be otherwise caused during manufacture by a gap height differential between the portion of the semiconductor die **706** coupled to at least one of the signal leads **712**, for example by bumps **710**, and the portion of the semiconductor die **706** coupled to the primary conductor **704**.

[0064] The detailed description set forth above, in connection with the appended drawings, is intended as a description of various configurations and is not intended to represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for providing a thorough understanding of the various concepts. It will be apparent to those skilled in the art, however, that these concepts may be practiced without these specific details. In some instances, well-known structures and components are shown in block diagram form in order to avoid obscuring such concepts.

[0065] Based on the teachings, one skilled in the art should appreciate that the scope of the present disclosure is intended to cover any aspect of the present disclosure, whether implemented independently of or combined with any other aspect of the present disclosure. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth. In addition, the scope of the present disclosure is intended to cover such an apparatus or method practiced using other structure, functionality, or structure and functionality in addition to,

or other than the various aspects of the present disclosure set forth. It should be understood that any aspect of the present disclosure may be embodied by one or more elements of a claim.

[0066] Although reference is made herein to particular materials, it is appreciated that other materials having similar functional and/or structural properties may be substituted where appropriate, and that a person having ordinary skill in the art would understand how to select such materials and incorporate them into embodiments of the concepts, techniques, and structures set forth herein without deviating from the scope of those teachings.

[0067] Various embodiments of the concepts, systems, devices, structures and techniques sought to be protected are described herein with reference to the related drawings. Alternative embodiments can be devised without departing from the scope of the concepts, systems, devices, structures and techniques described herein. It is noted that various connections and positional relationships (e.g., over, below, adjacent, etc.) are set forth between elements in the following description and in the drawings. These connections and/or positional relationships, unless specified otherwise, can be direct or indirect, and the described concepts, systems, devices, structures and techniques are not intended to be limiting in this respect. Accordingly, a coupling of entities can refer to either a direct or an indirect coupling, and a positional relationship between entities can be a direct or indirect positional relationship.

[0068] As an example of an indirect positional relationship, references in the present description to forming layer “A” over layer “B” include situations in which one or more intermediate layers (e.g., layer “C”) is between layer “A” and layer “B” as long as the relevant characteristics and functionalities of layer “A” and layer “B” are not substantially changed by the intermediate layer(s). The following definitions and abbreviations are to be used for the interpretation of the claims and the specification. As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” “contains” or “containing,” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a composition, a mixture, process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but can include other elements not expressly listed or inherent to such composition, mixture, process, method, article, or apparatus.

[0069] Additionally, the term “exemplary” is used herein to mean “serving as an example, instance, or illustration. Any embodiment or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. The terms “one or more” and “one or more” are understood to include any integer number greater than or equal to one, i.e., one, two, three, four, etc. The terms “a plurality” are understood to include any integer number greater than or equal to two, i.e., two, three, four, five, etc. The term “connection” can include an indirect “connection” and a direct “connection.”

[0070] References in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment can include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

[0071] For purposes of the description hereinafter, the terms “upper,” “lower,” “right,” “left,” “vertical,” “horizontal,” “top,” “bottom,” and derivatives thereof shall relate to the described structures and methods, as oriented in the drawing figures. The terms “overlying,” “atop,” “on top,” “positioned on” or “positioned atop” mean that a first element, such as a first structure, is present on a second element, such as a second structure, where intervening elements such as an interface structure can be present between the first element and the second element. The term “direct contact” means that a first element, such as a first structure, and a second element, such as a second

structure, are connected without any intermediary elements.

[0072] Use of ordinal terms such as “first,” “second,” “third,” etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements.

[0073] The terms “approximately” and “about” may be used to mean within $\pm 20\%$ of a target value in some embodiments, within $\pm 10\%$ of a target value in some embodiments, within $\pm 5\%$ of a target value in some embodiments, and yet within $\pm 2\%$ of a target value in some embodiments. The terms “approximately” and “about” may include the target value. The term “substantially equal” may be used to refer to values that are within $\pm 20\%$ of one another in some embodiments, within $\pm 10\%$ of one another in some embodiments, within $\pm 5\%$ of one another in some embodiments, and yet within $\pm 2\%$ of one another in some embodiments.

[0074] The term “substantially” may be used to refer to values that are within $\pm 20\%$ of a comparative measure in some embodiments, within $\pm 10\%$ in some embodiments, within $\pm 5\%$ in some embodiments, and yet within $\pm 2\%$ in some embodiments. For example, a first direction that is “substantially” perpendicular to a second direction may refer to a first direction that is within $\pm 20\%$ of making a 90° angle with the second direction in some embodiments, within $\pm 10\%$ of making a 90° angle with the second direction in some embodiments, within $\pm 5\%$ of making a 90° angle with the second direction in some embodiments, and yet within $\pm 2\%$ of making a 90° angle with the second direction in some embodiments.

[0075] As used herein, the term “determining” encompasses a wide variety of actions. For example, “determining” may include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), ascertaining and the like. Additionally, “determining” may include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Furthermore, “determining” may include resolving, selecting, choosing, establishing, and the like.

[0076] As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover: a, b, c, a-b, a-c, b-c, and a-b-c.

[0077] It is to be understood that the disclosed subject matter is not limited in its application to the details of construction and to the arrangements of the components set forth in the following description or illustrated in the drawings. The disclosed subject matter is capable of other embodiments and of being practiced and carried out in various ways. Also, it is to be understood that the phraseology and terminology employed herein are for the purpose of description and should not be regarded as limiting. As such, those skilled in the art will appreciate that the conception, upon which this disclosure is based, may readily be utilized as a basis for the designing of other structures, methods, and systems for carrying out the several purposes of the disclosed subject matter. Therefore, the claims should be regarded as including such equivalent constructions insofar as they do not depart from the spirit and scope of the disclosed subject matter.

[0078] The methods disclosed herein comprise one or more steps or actions for achieving the described method. The method steps and/or actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps or actions is specified, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims.

[0079] It is to be understood that the claims are not limited to the precise configuration and components illustrated above. Various modifications, changes, and variations may be made in the arrangement, operation, and details of the methods and apparatus described above without departing from the scope of the claims.

Claims

1. A current sensor integrated circuit (IC) package comprising: a lead frame comprising a primary conductor and signal leads, the lead frame having a lead frame surface; a semiconductor die having a die surface adjacent to the lead frame surface, the semiconductor die comprising a magnetic field sensing element supported by the semiconductor die, wherein the magnetic field sensing element is configured to sense a magnetic field associated with a current through the primary conductor and generate an output signal indicative of the current; and an insulation structure having first and second portions, wherein: the first portion is located between a first location on the die surface and the primary conductor; and the second portion is located between a second location on the die surface and a first one of the signal leads; wherein the insulation structure is configured to prevent a tilt of the semiconductor die.
2. The current sensor IC package of claim 1 wherein the first and second portions of the insulation structure comprise a single body of material.
3. The current sensor IC package of claim 2 wherein the single body comprises a T-shape.
4. The current sensor IC package of claim 1 wherein the insulation structure comprises a polymer film.
5. The current sensor IC package of claim 4 wherein the insulation structure further comprises an adhesive layer.
6. The current sensor IC package of claim 5 wherein the polymer film and the adhesive layer are provided as a tape.
7. The current sensor IC package of claim 6 wherein the tape extends beyond a periphery of the semiconductor die.
8. The current sensor IC package of claim 7 wherein the tape extends beyond the primary conductor adjacent to the signal leads by at least 0.4 mm.
9. The current sensor IC package of claim 1 wherein the insulation structure comprises one or more epoxy dots.
10. The current sensor IC package of claim 1 wherein the first portion comprises a polymer film and the second portion comprises an epoxy dot.
11. The current sensor IC package of claim 10 wherein the epoxy dot is substantially rectangular.
12. The current sensor IC package of claim 10 wherein the epoxy dot is substantially oval.
13. The current sensor IC package of claim 10 wherein the epoxy dot is polygonal.
14. The current sensor IC package of claim 10 wherein the epoxy dot is screen printed onto the first one of the signal leads.
15. The current sensor IC package of claim 10 wherein the epoxy dot is dimensioned to be within dimensions of the first one of the signal leads.
16. The current sensor IC package of claim 1 wherein the insulation structure is adapted to support the semiconductor die across one or both of a width or a length of the semiconductor die.
17. A method of manufacturing a current sensor integrated circuit (IC) package, the method comprising: providing a lead frame comprising a primary conductor and signal leads, the lead frame having a lead frame surface; positioning a semiconductor die having a die surface adjacent to the lead frame surface, wherein the semiconductor die supports a magnetic field sensing element, the magnetic field sensing element configured to sense a magnetic field associated with a current through the primary conductor and to generate an output signal indicative of the current; positioning a first portion of an insulation structure between a first location on the die surface and the primary conductor; and positioning a second portion of the insulation structure between a second location on the die surface and a first one of the signal leads; wherein the insulation structure is configured to prevent a tilt of the semiconductor die.
18. The method of claim 17 wherein the first and second portions of the insulation structure

comprise a single body of material.

19. The method of claim 18 wherein the single body comprises a T-shape.

20. The method of claim 17 wherein the insulation structure comprises a polymer film.

21. The method of claim 20 wherein the insulation structure further comprises an adhesive layer.

22. The method of claim 21 wherein the polymer film and the adhesive layer are provided as a tape.

23. The method of claim 22 wherein the tape extends beyond a periphery of the semiconductor die.

24. The method of claim 23 wherein the tape extends beyond the primary conductor adjacent to the signal leads by at least 0.4 mm.

25. The method of claim 17 wherein the insulation structure comprises one or more epoxy dots.

26. The method of claim 17 wherein the first portion comprises a polymer film and the second portion comprises an epoxy dot.

27. The method of claim 26 wherein the epoxy dot is substantially rectangular.

28. The method of claim 26 wherein the epoxy dot is substantially oval.

29. The method of claim 26 wherein the epoxy dot is polygonal.

30. The method of claim 26 wherein positioning the epoxy dot comprises screen printing the epoxy dot onto the first one of the signal leads.

31. The method of claim 26 further comprising dimensioning the epoxy dot to be within dimensions of the first one of the signal leads.

32. The method of claim 17 wherein the insulation structure is adapted to support the semiconductor die across one or both of a width or a length of the semiconductor die.

33. A current sensor integrated circuit (IC) package comprising: a lead frame comprising a primary conductor and signal leads, the lead frame having a lead frame surface; a semiconductor die having a die surface adjacent to the lead frame surface, the semiconductor die comprising a magnetic field sensing element supported by the semiconductor die, wherein the magnetic field sensing element is configured to sense a magnetic field associated with a current through the primary conductor and generate an output signal indicative of the current; and an insulation structure having first and second portions, wherein: the first portion is located between a first location on the die surface and the primary conductor; and the second portion is located between a second location on the die surface and a first one of the signal leads; wherein the insulation structure is configured to form a first gap height between the first location on the die surface and the primary conductor and a second gap height between the second location on the die surface and the first one of the signal leads.

34. The current sensor IC package of claim 33 wherein the first gap height and the second gap height are substantially equal.
